Sai Teja Mothukuri

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SUMMARY

AI/ML Engineer with 3+ years of experience designing and deploying machine learning, deep learning, multimodal AI, and transformer-based systems across cloud and edge environments. Experienced in recommendation systems, NLP, computer vision, reinforcement learning, and Generative AI applications, including RAG pipelines, LLM fine-tuning, and scalable ML platforms. Skilled in building end-to-end ML solutions from data engineering and experimentation to model deployment, MLOps, and production inference.

SKILLS

Machine Learning & AI: Deep Learning, Graph Neural Networks (GNNs), Reinforcement Learning, Recommendation Systems, NLP, Speech Recognition, Time-Series Forecasting, Feature Engineering, Hyperparameter Optimization, A/B Testing, Causal Inference, Predictive Modeling
LLM & Generative AI: LLM Fine-Tuning (LoRA, SFT), Retrieval-Augmented Generation (RAG), Hugging Face Transformers, LLaMA, Embeddings, Vector Databases, Semantic Search, Prompt Engineering, LangChain
Data Engineering & Distributed Systems: Python, SQL, PySpark, Apache Spark, Apache Kafka, ETL Pipelines, Real-Time Data Processing, Snowflake
Cloud & Infrastructure: AWS (S3, EC2, Lambda, SageMaker, RDS), Microservices Architecture, REST APIs, FastAPI
MLOps & Deployment: Docker, Kubernetes, MLflow, CI/CD (GitHub Actions), Model Monitoring, Drift Detection, Model Versioning
Data Analytics & Visualization: Power BI, Tableau

EXPERIENCE

AI/ML Engineer | Honeywell — **Aug 2025 – Present | Richmond, VA**

- Engineered multi-modal deep learning models using PyTorch, TensorFlow, and Hugging Face Transformers to integrate voice, image, and barcode data, achieving <200ms latency on edge devices while maintaining 98% accuracy in object and speech recognition tasks.
- Optimized on-device AI inference pipelines for ARM and NVIDIA Jetson architectures, reducing model memory footprint by 35% and enabling real-time processing of 5,000+ inputs/day across distribution center operations.
- Deployed NLP modules for workflow automation, leveraging transformer-based LLMs and RAG pipelines to extract task-relevant insights from worker instructions and manuals, improving operational guidance accuracy by 22%.
- Implemented reinforcement learning (RL) agents to enhance mobile workforce routing and task prioritization using real-time sensor and location data, resulting in a 15% increase in daily task completion efficiency.
- Constructed scalable vector databases and semantic retrieval pipelines to store multi-modal embeddings for rapid access, enabling sub-second responses for >15 million structured and unstructured records/month.
- Orchestrated MLOps pipelines on AWS and Kubernetes for continuous training, evaluation, and deployment, including 95+ automated retraining cycles/quarter, ensuring model drift detection and 99.5% uptime for edge agents.
- Applied model compression techniques including quantization and pruning to optimize multimodal and transformer-based models for edge deployment, improving inference throughput by 40% while reducing memory utilization and power consumption.

Machine Learning Scientist | Accenture — **Sep 2021 – Dec 2023 | India**

- Designed high-throughput advertising recommendation engine using Python, PySpark, and Spark Streaming, processing 20+ TB/month of clickstream and ad data, reducing ad targeting latency to <3 seconds per request while maintaining 99.9% availability.
- Formulated predictive models for ad performance using XGBoost, LightGBM, and LSTM-based models, incorporating 25+ behavioral and demographic features, achieving 15% uplift in click-through rate and 20% revenue growth across 95 KPIs
- Integrated and analyzed large-scale A/B experiments to evaluate recommendation model performance, applying causal inference and statistical hypothesis testing to validate CTR improvements across millions of user interactions.
- Developed graph-based recommendation models using Graph Neural Networks (GNNs) to capture user–item interaction networks, improving recommendation relevance and increasing click-through rates across large-scale advertising datasets.
- Architected real-time ingestion pipelines using Apache Kafka and Spark Streaming to integrate ad impressions, user behavior, and campaign data, enabling near-real-time feature updates and supporting low-latency recommendation inference across millions of requests.
- Built analytics dashboards using Power BI integrated with Snowflake, enabling monitoring of ad campaign performance, model KPIs, and engagement metrics, improving data-driven decision making for product teams.
- Established data analytics assistants using LangChain to query Snowflake advertising datasets through natural language interfaces, enabling analysts to generate campaign insights and performance summaries 40% faster across large-scale marketing data.
- Configured containerized model inference APIs using FastAPI and Docker to serve recommendation and prediction models, enabling real-time access for advertising platforms and supporting low-latency prediction requests per minute across campaign optimization workflows.

PROJECTS

Intelligent Real-Time Traffic Flow Prediction Using Multi-Modal Data | *Python, LSTM, MLOps*

- Programmed a real-time traffic prediction system using LSTM models, achieving 15–20% improved prediction accuracy and <2-second inference latency for smart city traffic optimization.
- Applied an end-to-end MLOps pipeline with Python, including data preprocessing, model training, and interactive dashboards, enabling actionable insights for traffic signal management.

Multi-Modal Cyberbullying Detection

- Developed an end-to-end multi-modal deep-learning system (PyTorch) classifying cyberbullying across text, image, and audio into six harm categories through a unified inference pipeline.
- Trained a bidirectional-LSTM text classifier reaching 83.4% accuracy (0.83 macro-F1) on 9,539 held-out samples and a VGG16 transfer-learning image model reaching 96.1% accuracy (0.96 macro-F1).
- Engineered the audio path with speech-to-text routed through the text model, with reproducible training – stratified splits, early stopping, and confusion-matrix evaluation.

EDUCATION

Masters in Computer Science | *University of Central Missouri, USA*